C_4-Free Subgraphs of the Hypercubes Q_6, Q_7, and Q_8: Odd Squares, Fully Frustrated Models, and Computational Structure

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This is a revision of a manuscript first circulated as arXiv:2603.29127 (v1–v4, March–May 2026), under the title "New Lower Bounds for C_4-Free Subgraphs of the Hypercubes Q_6, Q_7, and Q_8: Constructions, Structure, and Computational Method." That earlier version announced $\mathrm{ex}(Q_8, C_4) \geq 680$ and did not include the odd-square material of Section 5; see the Acknowledgements for the circumstances of the revision.

Abstract

We study C_4-free subgraphs of the hypercubes Q_6, Q_7, and Q_8, giving explicit lower bounds $\mathrm{ex}(Q_7, C_4) \geq 304$ and $\mathrm{ex}(Q_8, C_4) \geq 682$, together with a computational reproduction of the 132-edge lower-bound construction for $\mathrm{ex}(Q_6, C_4) = 132$, combined with the known upper bound of Harborth and Nienborg [8] (we did not independently verify ILP optimality for the upper bound within a practical runtime; see Section 8). After this manuscript was first circulated, a reader (S. Lai) identified an overlooked connection to a line of statistical-physics literature on fully frustrated hypercubes: under the standard correspondence between edge sets meeting every 4-cycle (*square*) an odd number of times and fully frustrated signed hypercubes, the 1979 construction of Derrida, Pomeau, Toulouse, and Vannimenus [5] already yields a 304-edge C_4-free subgraph of Q_7, and the D=8 ground state reported by Marinari, Parisi, and Ritort [10] yields a 682-edge C_4-free subgraph of Q_8 that improves our originally announced 680-edge construction. For Q_7, this value coincides with one of our own previously found 304-edge solutions, which we confirm independently satisfies the odd-square condition. For Q_8, we derived and machine-verified an explicit 682-edge witness attaining the bound MPR report attaining, using their canonical fully frustrated coupling (derivation documented in Section 5.3); this witness is not claimed to be a reconstruction of any particular configuration MPR may have used internally, since they did not publish one. This 682-edge witness is presented here as the paper's headline lower bound in place of the earlier 680-edge witness.

Writing $g(n)$ for the maximum size of an *odd-square* edge set (every square meets it in exactly one or three edges), we show $g(6) = 132$, $g(7) = 304$, and $g(8) = 682$

exactly. For $n = 7, 8$ the upper bound follows from the local-field argument of Derrida et al. [5] and the lower bound from explicit constructions; for $n = 6$ we instead combine an explicit odd-square construction already given in [5] itself (lower bound $g(6) \geq 132$) with the independent exact value $\mathrm{ex}(Q_6, C_4) = 132$ of Harborth and Nienborg [8], which bounds $g(6)$ from above since every odd-square set is C_4-free. This settles the odd-square subclass completely for $n = 6, 7, 8$ without resolving the unrestricted problem $\mathrm{ex}(Q_n, C_4)$, since a general C_4-free subgraph need not be odd-square.

An exhaustive audit of the 19 866 previously published 304-edge Q_7 solutions shows that only 389 of them lie in the odd-square subclass; the remaining 19 477 do not. The odd-square construction therefore does not subsume or fully explain the structural classification given here (one profile type, Type 18 with 101 solutions, is entirely odd-square; of the other 19, three (ranks 5, 7, and 10) are mixed, containing some odd-square and some non-odd-square solutions, and the remaining 16 contain no odd-square solutions at all): the 19 866 solutions' common structural core (degree sequence $\{4^{32}, 5^{96}\}$, spectral radius in $[4.78543, 4.79129]$, local maximality) and their classification into 20 dimension-profile types are independent results. Whether these 19 866 solutions exhaust all 304-edge C_4-free subgraphs of Q_7 remains open.

For Q_8, the reconstructed 682-edge odd-square witness is locally maximal in a strong sense: all 342 non-edges of Q_8 create at least 3 new C_4s upon addition (compared with the earlier 680-edge simulated-annealing Solution B, for which 2 of 344 non-edges created only 1 or 2 new C_4s; see Section 4 for both 680-edge solutions released). All C_4-free-construction bounds are witnessed by explicit edge-list constructions with machine-verifiable certificates – the sole exception being the odd-square lower bound $g(6) \geq 132$, which rests on Derrida et al.'s $D = 6$ bond configuration (Section 5.2), not reconstructed by us; the released 132-edge Q_6 witness (`q6_edges_132.jsonl`) is a genuine, machine-verified C_4-free construction establishing $\mathrm{ex}(Q_6, C_4) \geq 132$, but its square-intersection histogram {1:8, 2:44, 3:188} shows it is *not* itself an odd-square witness. For all other bounds, a complete, deterministic enumeration of all four-cycles (240 for Q_6, 672 for Q_7, 1 792 for Q_8) establishes C_4-freeness, and every *construction* certificate reported here (the edge sets themselves, their C_4-freeness, and the Q_8 odd-square condition) is reproducible by an independent, dependency-free verifier (`verify.py`), with the Q_7 odd-square membership of individual solutions separately reproducible via `audit_q7_odd_square.py`. The further structural and statistical claims reported below (spectral-radius ranges, pairwise Hamming extrema, Type-18 automorphism counts, non-edge violation distributions, and similar) were independently recomputed during this paper's preparation but are not packaged into a single bundled verifier; a reader wishing to check them would need to recompute them from the released edge lists using the definitions given in the text.

The original (Q_7, Q_8) constructions are found by a two-phase simulated annealing algorithm, described in full detail; the released search code computes an $\mathrm{Aut}(Q_n)$-diversified restart point before each trial but does not pass it into the phase-1 search, which always restarts from a fresh uniform-random edge set instead (see Section 7). For Q_6 we additionally give an ILP formulation whose feasible value matches the known exact value $\mathrm{ex}(Q_6, C_4) = 132$; we did not independently verify the ILP's optimality within a practical runtime, and the upper bound itself rests on the combinatorial proof of Harborth and Nienborg [8], which we reproduce on the lower (construction) side.

Edge lists, ILP files, the odd-square reconstruction and audit scripts, and source code are made available at https://github.com/minamominamoto/c4free-hypercube.

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Contents

1 Introduction

The n-dimensional hypercube Q_n has vertex set $\{0,1\}^n$, two vertices adjacent iff they differ in exactly one coordinate; it has 2^n vertices and $n \cdot 2^{n-1}$ edges. The problem of determining $\mathrm{ex}(Q_n, C_4) = \max\{|E(G)| : G \subseteq Q_n,\ G \text{ is } C_4\text{-free}\}$ was raised by Erdős [7] and remains open in general.

The best known asymptotic bounds are

$$\tfrac{1}{2}(n + \sqrt{n}) \cdot 2^{n-1} \;\le\; \mathrm{ex}(Q_n, C_4) \;\le\; (0.60318 + o(1)) \cdot n \cdot 2^{n-1}, \qquad (1)$$

where the lower bound (valid for $n = 4^r$) is due to Brass, Harborth, and Nienborg [3]. The upper bound is a statement about the edge Turán *density* $\pi_e = \limsup_n \mathrm{ex}(Q_n, C_4)/(n2^{n-1})$, not a pointwise inequality valid at every finite n; the constant 0.60318 is due to Baber [1], who used the semidefinite flag algebra method to improve the earlier bound of Thomason and Wagner [11]; the intermediate value 0.6068 was obtained independently by Baber [1] and by Balogh, Hu, Lidický, and Liu [2]. For general $n \geq 9$, the weaker bound $\frac{1}{2}(n + 0.9\sqrt{n}) \cdot 2^{n-1}$ applies [3]. Exact values have been determined for small n; Harborth and Nienborg [8] proved $\mathrm{ex}(Q_6, C_4) = 132$.

Our main results are:

Theorem 1. *(i)* $\mathrm{ex}(Q_7, C_4) \geq 304$.

(ii) $\mathrm{ex}(Q_8, C_4) \geq 682$.

Part (i) is not new: it follows from the 1979 construction of Derrida, Pomeau, Toulouse, and Vannimenus [5] (Section 5), and we present it here as an independently reproduced and machine-verified result rather than as a novel bound. Part (ii) improves our originally announced bound $\mathrm{ex}(Q_8, C_4) \geq 680$, but here too the bare existence claim is not new in substance: Marinari, Parisi, and Ritort [10] already report having exhibited a $D = 8$ ground state attaining Derrida et al.'s improved energy lower bound, and combined with the signed-graph/odd-square correspondence and edge-count–field-sum translation we give in Section 5, their published existence claim already implies the existence of a 682-edge odd-square subgraph of Q_8, independently of any witness search of our own. What Part (ii) contributes beyond that prior existence claim is: an explicit graph-theoretic translation of MPR's result into edge-count terms (Section 5.2); a publicly released, explicit, machine-verifiable 682-edge certificate, which MPR's paper does not supply (they report the result in energy units only, without an explicit spin configuration); and the structural analysis of that certificate (Proposition 11). The specific certificate we release attains the same bound value MPR report; its own provenance is recorded as a dated private-correspondence account rather than an independently reproducible computational log (see Section 5.3); we do not claim it reproduces any particular configuration MPR may have used internally. Both bounds are realised by explicit constructions (supplementary files `q7_edges_304.jsonl.part1--3` and `q8_odd_square_682.json`; the originally announced 680-edge Q_8 witness, `q8_edges_680.jsonl`, is retained as a simulated-annealing data point, see Section 4).

Within the *odd-square* subclass — edge sets meeting every square in an odd number of edges — the extremal value $g(n)$ is determined exactly for $n = 7, 8$:

Theorem 2. $g(6) = 132$, $g(7) = 304$, *and* $g(8) = 682$.

Theorem 2 is proved in Section 5. It does not resolve Theorem 1's unrestricted bounds, since $g(n) \leq \mathrm{ex}(Q_n, C_4)$ and a general C_4-free subgraph need not be odd-square: of the 19 866 published 304-edge Q_7 solutions, only 389 are odd-square (Section 6).

The paper is organised as follows. Section 2 describes the C_4 enumeration and verification procedure used throughout. Sections 3–4 present the originally announced Q_7 and Q_8 simulated-annealing constructions. Section 5 introduces the odd-square subclass, proves Theorem 2, and presents the reconstructed 682-edge Q_8 witness and its verification. Section 6 classifies the 19 866 Q_7 solutions found by their dimension profiles, yielding exactly 20 types, and reports the odd-square audit (389 of 19 866). Section 7 describes the two-phase simulated annealing algorithm. Section 8 reproduces $\mathrm{ex}(Q_6, C_4) = 132$

Table 1: Known values and lower bounds of $\mathrm{ex}(Q_n, C_4)$. The BHN column gives Brass–Harborth–Nienborg estimates from Eq. (1); that equation's tight form applies only for $n = 4^r$ and its weaker form only for $n \geq 9$ [3], so the $n = 5, 6$ entries are not covered by either stated form (marked n/a) and the $n = 7, 8$ entries are extrapolations of the $n \geq 9$ form *beyond* its stated domain, included for illustrative comparison only, not as a proven bound.

n	$\lvert E(Q_n)\rvert$	$\mathrm{ex}(Q_n, C_4)$	BHN (Eq. 1)	Source
4	32	24	24 (tight, $n = 4^1$)	Chung [4]
5	80	56	n/a (outside domain)	Emamy-K et al. [6]
6	192	132	n/a (outside domain)	Harborth–Nienborg [8]
7	448	$\geq$ **304**	≈ 300.2 (extrapol.)	Derrida et al. [5]
8	1024	$\geq$ **682**	≈ 674.9 (extrapol.)	witness for [10] bound

computationally on the lower-bound side, with an ILP formulation for the upper-bound side. Section 9 lists open problems.

2 C_4 Enumeration and Verification

Lemma 3 (C_4 enumeration in Q_n)**.** *Every four-cycle of Q_n is uniquely determined by a pair of coordinate directions $0 \leq i < j \leq n-1$ and a base vertex $b \in \{0,1\}^n$ with $b_i = b_j = 0$. Its four vertices are*

$$b, \quad b \oplus e_i, \quad b \oplus e_j, \quad b \oplus e_i \oplus e_j,$$

and the total number of four-cycles in Q_n is $\binom{n}{2} \cdot 2^{n-2}$. For $n = 6$: 240; for $n = 7$: 672; for $n = 8$: 1 792.

Verification algorithm. Given a candidate edge set $E \subseteq E(Q_n)$ stored as a hash set, the following procedure performs a complete, deterministic C_4-freeness check:

for each pair (i, j) with $0 \leq i < j \leq n-1$:
for each $b \in \{0,1\}^n$ with $b_i = b_j = 0$:
let $v_1 = b,\ v_2 = b \oplus e_i,\ v_3 = b \oplus e_j,\ v_4 = b \oplus e_i \oplus e_j$
if $\{v_1v_2,\ v_1v_3,\ v_2v_4,\ v_3v_4\} \subseteq E$: **return** VIOLATION
return C4-FREE

We applied this algorithm to all constructions and all 19 866 solutions of Q_7; every call returned C4-FREE.

3 The Q_7 Construction

We constructed a C_4-free subgraph of Q_7 on 304 edges by penalty-based simulated annealing (Section 7), then certified it by Lemma 3's algorithm.

Proposition 4. *The 304-edge construction has the following properties:*

(i) Degree sequence: 32 vertices of degree 4 and 96 of degree 5 (degree sum $608 = 2 \cdot 304$).

(ii) *Spectral radius $\lambda_1 \approx 4.787$ (a genuine computed quantity for this construction); $\lambda_n \approx -4.787$, i.e. $\lambda_1 + \lambda_n = 0$ to machine precision, which is automatic for any subgraph of the bipartite Q_7 and serves here only as a sanity check on the eigenvalue computation, not as an independent structural finding.*

(iii) $\mathrm{Tr}(A^4) = 5216$, *attaining the C_4-free trace equality* $\sum_i \deg(i)^2 + \sum_{uv\in E}(\deg u + \deg v) - 2|E|$.

(iv) *Local maximality: every non-edge of Q_7 creates a C_4 when added.*

Through 19 866 reported simulated-annealing runs (Section 6.3; see Section 7 for what is and is not independently verifiable about this search history) we found 19 866 C_4-free subgraphs of Q_7 on 304 edges with pairwise distinct edge sets (not quotiented by $\mathrm{Aut}(Q_7)$), all sharing the structural properties stated in Proposition 14.

4 The Q_8 Construction

This section documents the originally announced Q_8 witness, obtained independently of the odd-square subclass by the same simulated-annealing method as the Q_7 construction of Section 3. It has since been superseded as the paper's headline lower bound by the 682-edge odd-square reconstruction of Section 5; it is retained here as an independent structural data point and as a demonstration of the simulated-annealing method on its own terms.

Simulated annealing produced two C_4-free subgraphs of Q_8 on 680 edges (both released in `q8_edges_680.jsonl`), each certified by exhaustive enumeration of all 1 792 four-cycles. We refer to them as Solution A (first line of the file) and Solution B (second line); the two are structurally distinct and are kept separate below, since an earlier draft of this section inadvertently described properties of Solution A and Solution B as if they belonged to a single construction.

Proposition 5. *Both 680-edge solutions are C_4-free and locally maximal (every non-edge of Q_8 creates at least one C_4 upon addition), with edge density $680/1024 = 66.4\%$ (compared with $304/448 = 67.9\%$ for the Q_7 construction) and $\lambda_1 + \lambda_n = 0$ (automatic for any subgraph of the bipartite Q_8; not by itself a distinguishing structural feature). They differ as follows:*

(i) *Solution A: degree sequence $\{4^8, 5^{160}, 6^{88}\}$ (degree sum $1360 = 2 \cdot 680$); square-intersection histogram $\{1:308, 3:1484\}$ over the 1 792 squares (incidentally odd-square, though not constructed as such); non-edge violation distribution $\{2:3, 3:48, 4:144, 5:136, 6:13\}$ (minimum 2 new C_4s per non-edge).*

(ii) *Solution B: degree sequence $\{4^7, 5^{162}, 6^{87}\}$ (degree sum $1360 = 2 \cdot 680$); square-intersection histogram $\{1:302, 2:12, 3:1478\}$ (not odd-square: 12 squares meet it in exactly 2 edges); non-edge violation distribution $\{1:1, 2:1, 3:49, 4:153, 5:124, 6:16\}$ (minimum 1 new C_4, attained by exactly one non-edge).*

Remark 6. Brass–Harborth–Nienborg's tight bound $\frac{1}{2}(n + \sqrt{n}) \cdot 2^{n-1}$ applies only for $n = 4^r$, and their weaker bound $\frac{1}{2}(n + 0.9\sqrt{n}) \cdot 2^{n-1}$ is stated only for $n \geq 9$ (Eq. (1), [3]); neither covers $n = 8$. Evaluating the $n \geq 9$ form at $n = 8$ nonetheless gives ≈ 674.9; both 680-edge constructions and the 682-edge odd-square reconstruction of Section 5 exceed this extrapolated figure (by 5.1 and 7.1 edges respectively), but we do not claim this as a comparison against a proven bound.

4.1 Local optimality of the 680-edge constructions

Solution B is locally maximal in the weakest possible sense among the two: among its 344 non-edges, exactly 1 creates exactly 1 new C_4; exactly 1 creates exactly 2; the remaining 342 create 3 or more (Proposition 5(ii)). Solution A's non-edges all create at least 2 new C_4s.

Over 1 076 reported simulated-annealing trials targeting 681 edges, the minimum C_4 violation count observed was 1 (never 0), where the violation count denotes the number of four-cycles fully contained in the candidate edge set. This was originally reported as computational evidence for the conjecture $\mathrm{ex}(Q_8, C_4) = 680$; that conjecture is withdrawn, since the odd-square reconstruction of Section 5 gives $\mathrm{ex}(Q_8, C_4) \geq 682 > 680$. Since a stochastic annealing run does not define a graph-theoretic neighbourhood search (no fixed radius or reachability guarantee is established), the 681-edge search failure is evidence only about the behaviour of the stated heuristic across 1 076 independent fresh-random-restart trials targeting 681 edges (Section 7), not a mathematical statement about $\mathrm{ex}(Q_8, C_4)$ or about any formally defined neighbourhood of the 680-edge solutions.

Table 2 compares the Q_7 construction with Solution B (the 680-edge solution whose local-optimality numbers are discussed above); see Table 3 in Section 5 for the odd-square reconstructions.

Table 2: Structural comparison of the Q_7 construction and Q_8 Solution B (see Proposition 5 for Solution A).

Parameter	Q_7 (304 edges)	Q_8 Solution B (680 edges)
Total edges $\lvert E(Q_n)\rvert$	448	1024
Achieved edges	304	680
Edge density	67.9%	66.4%
Min degree	4	4
Max degree	5	6
Degree sequence	$\{4^{32}, 5^{96}\}$	$\{4^7, 5^{162}, 6^{87}\}$
BHN (Eq. 1), extrapolated	≈ 300.2	≈ 674.9
Improvement	$+3.8$	$+5.1$

5 The Odd-Square Subclass

5.1 Definition and equivalence to fully frustrated hypercubes

Definition 7. An edge set $E \subseteq E(Q_n)$ is *odd-square* if every square (four-cycle) of Q_n meets E in an odd number of edges, i.e. one or three of its four edges lie in E. Write $g(n) = \max\{|E| : E \text{ is odd-square in } Q_n\}$.

Every odd-square edge set is C_4-free, since no square can meet it in all four edges; hence $g(n) \leq \mathrm{ex}(Q_n, C_4)$. The odd-square condition is exactly the condition studied under a different name in the statistical-physics literature on *fully frustrated hypercubes*: assign to each edge e a sign $\sigma(e) = +1$ if $e \in E$ and $\sigma(e) = -1$ otherwise; the square condition becomes "the product of signs around every square is -1", i.e. every square is frustrated.

Lemma 8. *Let $\sigma_1, \sigma_2 : E(Q_n) \to \{\pm 1\}$ both have product -1 on every square. Then there is a function $t : \{0,1\}^n \to \{\pm 1\}$ with $\sigma_2(x,y) = \sigma_1(x,y)\, t(x)t(y)$ for every edge (x,y).*

Proof. Let $\tau = \sigma_1\sigma_2$ (pointwise product). On every square C,

$$\prod_{e\in C} \tau(e) = \Big(\prod_{e\in C} \sigma_1(e)\Big)\Big(\prod_{e\in C} \sigma_2(e)\Big) = (-1)(-1) = +1.$$

The map sending a cycle C to $\prod_{e\in C} \tau(e) \in \{\pm 1\}$ is a group homomorphism from the cycle space of Q_n (over GF(2), under symmetric difference) to $\{\pm 1\}$, since edges shared by two cycles cancel in their symmetric difference and likewise cancel in the product. The squares of Q_n generate its cycle space (a standard fact for hypercubes: every cycle is a symmetric difference of squares), so this homomorphism is determined by its values on the squares; since it is $+1$ on every square, it is $+1$ on every cycle. A signature with product $+1$ on every cycle of a connected graph is a *coboundary*: $\tau(x,y) = t(x)t(y)$ for some $t : V \to \{\pm 1\}$ (the classical balance theorem for signed graphs, e.g. via Harary's theorem). Solving for σ_2 gives the claim. $\square$

Lemma 8 says any two odd-square signatures are related by a *vertex switching* t. For a *fixed* coupling J, letting the spin configuration $s : V(Q_n) \to \{\pm 1\}$ range over all 2^{2^n} choices (one sign per vertex of Q_n, which itself has 2^n vertices) produces exactly the signatures $\sigma_s(x,y) = J(x,y)s(x)s(y)$ in the switching class of J (writing $s = t \cdot s_0$ for a fixed reference s_0 realises every switching t); this is the operational fact we use, not a claim that different s give the same energy under fixed J (they generally do not – that variation is exactly what is being optimised). The energy-preserving gauge transformation in the physical sense instead acts on *both* J and s together, $J(x,y) \mapsto J(x,y)t(x)t(y)$ and $s(x) \mapsto t(x)s(x)$, under which $J(x,y)s(x)s(y)$ is unchanged edge-by-edge [10]. Consequently, $g(n)$ — defined as a maximum over *all* odd-square edge sets — equals the maximum of $|E|$ over the switching class of any one fixed fully-frustrated signature, in particular the reference coupling J used in Section 5.3's witness; optimising that one fixed J (or, for $D = 8$, exhibiting a ground state attaining Derrida et al.'s improved energy lower bound for it, as Marinari–Parisi–Ritort report [10]) therefore does optimise over the class defining $g(n)$, not merely over a possibly-proper subclass of it.

Under the further correspondence between signed hypercubes and Ising spin configurations $s : \{0,1\}^n \to \{\pm 1\}$ via $\sigma(x,y) = J(x,y)\, s(x)s(y)$ for a fixed reference coupling J with every square frustrated, maximising $|E|$ (i.e. maximising the number of *positive* edges, equivalently minimising the number of negative edges) is exactly the problem of minimising the ground-state energy of the fully frustrated Ising hypercube studied by Derrida, Pomeau, Toulouse, and Vannimenus [5] and, for $D = 8$, by Marinari, Parisi, and Ritort [10]: the ground-state energy is (up to an affine rescaling) the number of negative edges, $|E(Q_n)| - |E|$. In the language of signed graphs, $|E(Q_n)| - g(n)$ is the *frustration index* of the fully frustrated signature on Q_n. Laplante, Naserasr, Sunny, and Wang [9] study this frustration index directly (in the context of Huang's proof of the sensitivity conjecture) and explicitly establish a connection between it and Erdős's question on the largest C_4-free subgraph of the hypercube, giving lower and upper bounds on the frustration index of the fully frustrated signature that match when n is a power of 4; our $g(6), g(7), g(8)$ results above sharpen this connection to exact values at $n = 6, 7, 8$ specifically. This sign convention matches the one used by the reconstruction script `generate_q8_682.py` (Section 5.3), where E is exactly the set of edges with $J(x, \mathrm{dim})\, s(x)s(y) = +1$; for a vertex v of degree $\deg_E(v)$ in E, we call $h(v) = 2\deg_E(v) - n$ its *local field* under this convention (matching the local-field histogram reported in Proposition 11 below). We do not reproduce Derrida et al.'s derivation of the local-field lower bound here; Section 5.2 cites it as an external result, established for $D \leq 7$ by explicit construction and used by Marinari–Parisi–Ritort's $D = 8$ construction, which we reconstruct and verify independently in Section 5.3 without relying on the bound's derivation.

This connection was brought to our attention, after the original circulation of this manuscript, by S. Lai (see Acknowledgements), who identified the relevance of [5, 10] and located the improved Q_8 result.

5.2 Exact values for $n = 6, 7, 8$

Derrida et al. [5] give a local-field lower bound on the number of negative (frustrated, in their terminology) edges, tight enough to determine the fully frustrated hypercube's ground-state energy exactly for $D \leq 7$ via an explicit recursive construction, and conjectured it remains tight for $D \geq 8$. For $D = 6$, they give this explicit construction directly: "this configuration for H_6 contains 60 negative bonds (out of a total of 192)" [5, p. 622], i.e. $192 - 60 = 132$ positive (unfrustrated) edges, with the construction guaranteed fully frustrated (every square meeting it in an odd number of negative edges) by their general recursive method. Under the correspondence of Section 5 this is an explicit odd-square witness with 132 edges, so $g(6) \geq 132$; combined with $g(6) \leq \mathrm{ex}(Q_6, C_4) = 132$ [8], this gives $g(6) = 132$. We did not reconstruct this $D = 6$ configuration ourselves (unlike the $D = 7, 8$ cases below); the value 132 rests on the quoted primary-source count together with the independently proved exact value from [8], not on our own machine verification of the specific bond configuration.

Lemma 9 (Self-contained bound for $n = 7, 8$)**.** *For any odd-square edge set E in Q_n, writing U for one part of the bipartition ($|U| = 2^{n-1}$) and $h(v) = 2\deg_E(v) - n$ as before, every $h(v)$ with $v \in U$ is an integer of the same parity as n, and*

$$\sum_{v \in U} h(v)^2 = n \cdot 2^{n-1}.$$

Proof. For $v \in U$ and direction $i \in \{0, \dots, n-1\}$ write $s_i(v) = \sigma(v, v \oplus e_i) \in \{\pm 1\}$, so $h(v) = \sum_i s_i(v)$ and $h(v)^2 = n + \sum_{i \neq j} s_i(v) s_j(v)$. Fix $i \neq j$ and consider, for $v \in U$, the square on $\{v, v \oplus e_i, v \oplus e_j, z\}$ where $z = v \oplus e_i \oplus e_j \in U$. Its four edges have signs $s_i(v)$, $s_j(v)$, and (computed from z's own perspective, since a sign is a single scalar attached to the edge) $s_j(z)$, $s_i(z)$. The odd-square condition means their product is -1: $s_i(v)s_j(v) \cdot s_j(z)s_i(z) = -1$. Writing $X = s_i(v)s_j(v)$ and $Y = s_i(z)s_j(z)$, both lie in $\{\pm 1\}$ and $XY = -1$ forces $X = -Y$, i.e. $s_i(v)s_j(v) + s_i(z)s_j(z) = 0$. The map $v \mapsto z = v \oplus e_i \oplus e_j$ is a fixed-point-free involution on U (as $e_i \oplus e_j \neq 0$), so it partitions U into pairs $\{v, z\}$ on each of which the two corresponding terms cancel; summing over all such pairs gives $\sum_{v \in U} s_i(v)s_j(v) = 0$ for every fixed $i \neq j$. Summing over all ordered pairs $i \neq j$ and adding the $n \cdot |U|$ diagonal contribution gives $\sum_{v \in U} h(v)^2 = n \cdot 2^{n-1}$. □

This proof uses only that *every* square has product -1 (the odd-square condition), so it is sufficient for the identity to hold; it is not necessary. Testing against all 19 866 published 304-edge Q_7 solutions – odd-square and non-odd-square alike – shows every one of them satisfies $\sum_{v \in U} h(v)^2 = 448$, including the 19 477 that are not odd-square. The identity's failure for the non-odd-square Q_8 Solution B of Section 4 ($1016 \neq 1024$) is therefore not explained by odd-squareness as such: Solution B has an asymmetric degree distribution between U and its complement ($\{4^3, 5^{82}, 6^{43}\}$ vs. $\{4^4, 5^{80}, 6^{44}\}$), whereas Solution A, all 19 866 Q_7 solutions, and the 682-edge witness all have symmetric U/complement degree distributions. This symmetry is not itself implied by odd-squareness, however: a small odd-square counterexample in Q_3 (vertices $0, \dots, 7$, $E = \{(0,2), (2,6), (3,7), (4,5), (4,6), (6,7)\}$, whose six squares meet E in $1, 1, 3, 1, 3, 3$ edges) has U-degree multiset $\{1, 1, 1, 3\}$ against $\{0, 2, 2, 2\}$ on the complement, while still satisfying $\sum_{v \in U} h(v)^2 = \sum_{v \in U^c} h(v)^2 = 12 = 3 \cdot 2^2$ on each side separately. We do not characterise the identity's full necessary-and-sufficient condition here.

Lemma 10 (Nonnegativity at the maximum)**.** *If E is odd-square in Q_n with $|E|$ maximum (i.e. $|E| = g(n)$), then $h(v) \geq 0$ for every $v \in \{0,1\}^n$.*

Proof. Suppose $h(v) < 0$ for some v. Flipping the spin at v (replacing $s(v)$ by $-s(v)$, all other spins unchanged) flips $\sigma(v, w)$ for every edge (v, w) incident to v and leaves every other edge's sign unchanged. Every square containing v has exactly two edges incident to v; flipping both leaves their product, and hence the whole square's sign product, unchanged, so odd-squareness is preserved everywhere. At v itself, the positive-edge count changes from $p = (n + h(v))/2$ to $q = (n - h(v))/2$, a change of $q - p = -h(v) > 0$; no edge other than the n edges incident to v changes sign, so $|E|$ increases by exactly $-h(v) > 0$. This contradicts maximality of $|E|$, so no such v exists. □

This is a corollary of, not an alternative to, Lemma 8: the argument uses only that flipping one vertex's spin preserves odd-squareness (immediate from the switching correspondence) and counts the resulting change in $|E|$ directly.

Given Lemmas 9 and 10, $|E| = (n \cdot 2^{n-1} + \sum_{v \in U} h(v))/2$ is bounded above by relaxing to the integer optimisation problem of maximising $\sum_{v \in U} h(v)$ subject to the *fixed* sum of squares $\sum_{v \in U} h(v)^2 = n \cdot 2^{n-1}$ over the 2^{n-1} nonnegative integers $h(v)$ of fixed parity (we use only that every realisable h-vector satisfies these constraints, not the converse that every integer vector satisfying them is realisable as an odd-square signature; the bound obtained is therefore an upper bound on $|E|$, tight whenever a matching witness is exhibited). Both cases $n = 7, 8$ admit a short, fully elementary and self-contained upper bound on $\sum_{v \in U} h(v)$, without appeal to DPTV's Table I:

$n = 7$ **(odd $h \geq 0$).** For every odd integer $h \geq 0$, $(h-1)(h-3) \geq 0$, i.e. $h^2 \geq 4h - 3$, with equality exactly at $h \in \{1, 3\}$. Summing over $v \in U$ ($|U| = 64$) and using $\sum_v h(v)^2 = 448$:

$$448 = \sum_{v \in U} h(v)^2 \geq 4 \sum_{v \in U} h(v) - 3 \cdot 64, \quad \text{so} \quad \sum_{v \in U} h(v) \leq \frac{448 + 192}{4} = 160.$$

The released 304-edge witness attains $\sum_{v \in U} h(v) = 160$ exactly (Section 6.4), so this bound is tight and $g(7) = (448 + 160)/2 = 304$ is established without external input.

$n = 8$ **(even $h \geq 0$).** For every even integer $h \geq 0$, $(h-2)(h-4) \geq 0$, i.e. $h^2 \geq 6h - 8$, with equality exactly at $h \in \{2, 4\}$. Summing over $v \in U$ ($|U| = 128$) and using $\sum_v h(v)^2 = 1024$:

$$1024 = \sum_{v \in U} h(v)^2 \geq 6 \sum_{v \in U} h(v) - 8 \cdot 128, \quad \text{so} \quad \sum_{v \in U} h(v) \leq \frac{1024 + 1024}{6} = 341.\overline{3}.$$

Since $\sum_{v \in U} h(v)$ is a sum of even integers it is itself even, so $\sum_{v \in U} h(v) \leq 340$. The released 682-edge witness attains $\sum_{v \in U} h(v) = 340$ exactly, with h-distribution on U equal to {0:1, 2:84, 4:43} (independently recomputed from the released edge list); this three-value distribution is not two-valued, consistent with the equality cases $h \in \{2, 4\}$ of the inequality above together with the single $h = 0$ vertex, whose contribution the bound already prices in without requiring it to vanish. This bound is therefore also tight, giving $g(8) = (1024 + 340)/2 = 682$ without external input.

Both upper bounds $g(7) \leq 304$ and $g(8) \leq 682$ are thus established directly from Lemma 9 and elementary inequalities, independently of DPTV's Table I; we retain the citations to Derrida et al. [5] and Marinari–Parisi–Ritort [10] for historical priority and for the connection that prompted this section (Section 5.2), not because the upper-bound proof depends on them.

Marinari, Parisi, and Ritort [10] report having exhibited, by simulated annealing, an explicit $D = 8$ ground state attaining Derrida et al.'s improved energy lower bound – the same bound value $\sum_{v \in U} h(v) = 340$ established directly above. Their paper reports the result in energy units only, without an explicit spin configuration or its translation into edge-count terms; we supply that translation and an explicit, machine-verified 682-edge certificate in Section 5.3 below. Under the correspondence of Section 5, this historical attainment report at $D = 8$ corresponds to $g(8) = 682$, matching what was already established above from Lemmas 9 and 10 together with the integer/parity constraints and elementary inequalities; the explicit witness realising it is due to the present reconstruction, prompted by S. Lai's identification of the connection to [5, 10]. Together with the $D \leq 7$ case this gives Theorem 2: $g(6) = 132$, $g(7) = 304$, and $g(8) = 682$.

5.3 Verification

We reconstructed explicit witnesses for both values and machine-verified them independently of [5, 10].

Q_7, $g(7) = 304$. The 304-edge odd-square value coincides with our originally announced Q_7 lower bound (Theorem 1(i)); of the 19 866 solutions we found by simulated annealing (Section 3), 389 are themselves odd-square witnesses of $g(7) = 304$ (Section 6), independently confirming reachability of this value without reference to [5].

Q_8, $g(8) = 682$. We obtained a 682-edge odd-square witness for Q_8 from an explicit 256-bit spin configuration together with the canonical fully frustrated coupling $J(x, \dim) = (-1)^{\mathrm{popcount}(x \,\&\, (2^{\dim} - 1))}$ (one coupling convention realising a fully frustrated signature; see Section 5 supplementary script `generate_q8_682.py`). The script is dependency-free (Python standard library only), recomputes the edge set from the spin configuration, and asserts every structural quantity below against the recomputed data before writing the certificate; it does not take any of these quantities as input. *Provenance of the spin configuration.* The 256-bit configuration was derived by the author on 17–18 August 2026, using the canonical fully frustrated coupling of Marinari, Parisi, and Ritort [10], as recorded in dated correspondence with S. Lai, who independently reconstructed the identical 682-edge set from the shared spin vector and cross-checked all 1,792 squares, the degree distribution, and the local-field distribution before this material was incorporated into the manuscript. This provenance record is a first-person, dated account (the correspondence itself), not an independently reproducible computational log: no search seed, intermediate search records, or derivation script for the spin configuration itself is included in the released materials, only the scripts that regenerate and verify the edge set *from* the given spin configuration (Section 5). We do not claim this configuration is identical to any specific configuration MPR may have used internally, since they did not publish one; it is an independently derived witness attaining the same bound.

Proposition 11. *The witness edge set satisfies:*

(i) 682 *edges, spanning all* 256 *vertices of* Q_8*.*

(ii) *Square-intersection histogram* $\{1\!:\!301,\ 3\!:\!1491\}$ *over all* 1 792 *squares (odd-square condition holds everywhere; zero squares meet the edge set in* 0*,* 2*, or* 4 *edges).*

(iii) *Degree sequence* $\{4^2,\ 5^{168},\ 6^{86}\}$*; degree sum* $1364 = 2 \cdot 682$*.*

(iv) *Local-field histogram* $\{0\!:\!2,\ 2\!:\!168,\ 4\!:\!86\}$*, where the local field at a vertex is (positive degree)* $-$ *(negative degree) under the signed-graph correspondence above.*

(v) *Local maximality: every one of the* 342 *non-edges of* Q_8 *creates at least* 3 *new* C_4*s upon addition (distribution* $\{3\!:\!41,\ 4\!:\!159,\ 5\!:\!120,\ 6\!:\!22\}$*), a strictly stronger local-maximality margin than the originally announced 680-edge Solution B of Section 4, for which 2 of 344 non-edges created only 1 or 2 new* C_4*s.*

All quantities in Proposition 11 were recomputed independently of the reconstruction script by direct enumeration of all 1 792 squares against the released edge list, using the strict definition of C_4-freeness from Lemma 3 (no square may have all four edges present); this independent check found zero violations and is fully reproducible from the released files alone. The reconstruction, its SHA-256 checksum, and an independent audit are additionally reported to have been cross-checked against S. Lai's separately produced implementation, with full agreement; this cross-check is recorded as a dated correspondence account (as in Section 5.3), not as an independently verifiable artefact, since the correspondence, Lai's implementation, and its output are not included in the released materials.

Table 3 summarises the odd-square reconstructions alongside the corresponding simulated-annealing witnesses of Sections 3 and 4.

Table 3: Odd-square reconstructions compared with the simulated-annealing witnesses of Sections 3–4.

Parameter	Q_7 odd-square	Q_8 odd-square
Edges (value of $g(n)$)	304	682
Source	one of 389 SA sols.	witness for [10] bound
Non-odd-square SA witness (this paper)	—	680 (Solution B, 4)
Improvement over that witness	—	+2

6 Structural Classification of Q_7 Solutions

6.1 Dimension profiles and the twenty types

Definition 12 (Dimension profile)**.** For $G \subseteq Q_7$, let $e_i(G) = |\{uv \in E(G) : u \oplus v = 2^i\}|$. The *dimension profile* is the sorted tuple $\pi(G) = (e_{\sigma(0)} \geq \cdots \geq e_{\sigma(6)})$.

The dimension profile is invariant under coordinate permutations (a subgroup of $\mathrm{Aut}(Q_7)$ of order $7! = 5040$), giving a coarse but computable classification.

Table 4: The 20 dimension-profile types of 304-edge C_4-free subgraphs of Q_7, sorted by decreasing frequency. H: Shannon entropy of normalised profile (bits).

Rank	Profile π	Solutions	H	λ_1 (mean)
1	$[44, 44, 44, 44, 43, 43, 42]$	3155	2.8072	4.787
2	$[45, 45, 45, 43, 43, 42, 41]$	2913	2.8065	4.788
3	$[45, 45, 45, 43, 43, 43, 40]$	2200	2.8063	4.787
4	$[44, 44, 44, 44, 44, 43, 41]$	2116	2.8069	4.787
5	$[46, 46, 46, 42, 42, 42, 40]$	1756	2.8053	4.787
6	$[46, 46, 46, 42, 42, 41, 41]$	1181	2.8054	4.789
7	$[44, 44, 44, 44, 44, 44, 40]$	1085	2.8066	4.787
8	$[45, 45, 45, 43, 42, 42, 42]$	1064	2.8066	4.788
9	$[45, 45, 44, 43, 43, 43, 41]$	974	2.8067	4.787
10	$[44, 44, 44, 44, 44, 42, 42]$	931	2.8070	4.787
11	$[47, 47, 47, 41, 41, 41, 40]$	902	2.8037	4.788
12	$[45, 45, 43, 43, 43, 43, 42]$	439	2.8069	4.788
13	$[45, 44, 44, 43, 43, 43, 42]$	307	2.8070	4.788
14	$[46, 45, 45, 42, 42, 42, 42]$	236	2.8063	4.789
15	$[44, 44, 44, 43, 43, 43, 43]$	163	2.8073	4.787
16	$[47, 47, 44, 43, 41, 41, 41]$	153	2.8050	4.788
17	$[46, 46, 44, 42, 42, 42, 42]$	107	2.8062	4.789
18	$[48, 48, 48, 40, 40, 40, 40]$	101	2.8014	4.788
19	$[46, 46, 43, 43, 42, 42, 42]$	44	2.8063	4.788
20	$[46, 46, 45, 42, 42, 42, 41]$	39	2.8058	4.788
	Total	**19866**		

Proposition 13. *Among the* 19 866 *solutions, the dimension profile takes exactly* **20** *distinct values.*

Table 4 lists all 20 types with frequencies and structural invariants.

Observations: (1) λ_1 lies in $[4.787, 4.789]$ across all 20 types. (2) H ranges from 2.801 to 2.807, all near $\log_2 7 \approx 2.807$. (3) All 101 Type-18 solutions have a genuine $\mathrm{Aut}(Q_7)$-automorphism (coordinate permutation combined with a bitwise XOR, i.e. the full affine group $\mathrm{Aut}(Q_7) \cong (\mathbb{Z}_2)^7 \rtimes S_7$, not merely a coordinate permutation) that nontrivially permutes at least two of the three tied 48-edge directions; of these, 46/101 have a genuine order-3 automorphism cyclically permuting all three directions. (An earlier check that considered only pure coordinate permutations, without combining them with the XOR/translation part of $\mathrm{Aut}(Q_7)$, incorrectly found no such automorphisms in a 20-solution sample; that check was in error and has been redone exhaustively over all 101 solutions.)

6.2 Shared structural properties

Proposition 14. *Every solution in the* 19 866 *satisfies:*

(i) Degree sequence $\{4^{32}, 5^{96}\}$.

(ii) λ_1 *ranges over* $[4.78543, 4.79129]$ *across all* 19 866 *solutions (exhaustively computed; not a single shared three-decimal value), with* $\lambda_1 + \lambda_n = 0$ *throughout, automatic for*

any subgraph of the bipartite Q_7 (sanity check, not an independent finding). The type-wise means of Table 4 round to 4.787–4.789*.*

(iii) $\mathrm{Tr}(A^4) = 5216$ *(C_4-free trace equality).*

(iv) Local maximality: no edge of Q_7 can be added without creating a C_4.

(v) Every edge of Q_7 appears in at least one solution (verified by taking the union of the edge sets over all 19 866 *solutions).*

6.3 Solution landscape

Pairwise Hamming distances $d_H(G, G') = |E(G) \triangle E(G')|$, computed exhaustively over all $\binom{19\,866}{2} \approx 1.97 \times 10^8$ pairs, range from **6** (attained by a pair of profile-type-1 and profile-type-2 solutions, differing by just 3 edges each way) to **274**. A random sample of 5 000 pairs gives a mean ≈ 195.4 and median 196, both stable across resampling; the sample-based min and max are not, and should not be read as close to the true extremes above.

Over 1 076 reported trials targeting 305 edges, the minimum C_4 violation count was 2 (never 0).

Conjecture 15. $\mathrm{ex}(Q_7, C_4) = 304$*.*

6.4 Odd-square audit

We audited all 19 866 solutions of Section 3 against the odd-square condition of Section 5 (every square meets the solution in exactly one or three edges). Exactly 389 solutions are odd-square; the remaining 19 477 are not. Table 5 gives the dimension-profile breakdown of the 389 odd-square solutions; each belongs to one of only 4 of the 20 profile types of Table 4 (ranks 5, 7, 10, and 18 there).

Table 5: Dimension-profile breakdown of the 389 odd-square solutions among the 19 866 published 304-edge Q_7 solutions.

Dimension profile (sorted)	Count
$\{44, 44, 44, 44, 44, 44, 40\}$	127
$\{44, 44, 44, 44, 44, 42, 42\}$	83
$\{48, 48, 48, 40, 40, 40, 40\}$	101
$\{46, 46, 46, 42, 42, 42, 40\}$	78
Total odd-square	389
Total non-odd-square	19 477

Since the odd-square subclass accounts for only $389/19\,866 \approx 2.0\%$ of the published solutions, the shared structural core of Proposition 14 and the 20-type classification of Section 6.1 are not fully explained or subsumed by the odd-square construction: the great majority of 304-edge C_4-free subgraphs of Q_7 found here lie outside the fully frustrated switching class of [5]. The one exception is Type 18 ($[48, 48, 48, 40, 40, 40, 40]$, 101 solutions), all of which are odd-square; the other three types containing odd-square solutions (ranks 5, 7, and 10) are mixed, containing both odd-square and non-odd-square members.

7 Computational Method

The successful outcomes described in this section (the released edge sets themselves) are machine-verifiable certificates: any reader can confirm C_4-freeness and the reported edge counts directly from the supplied data with `verify.py`, independently of how the search was run. The quantitative claims about the *search process* that produced them — "19 866 independent runs", "1 076 independent trials" at 305 and at 681 edges, and the resampling statistics of Section 6.3 — describe unarchived search history. The search program itself, `c4free_sa.py`, is released alongside this manuscript, but the specific run seeds, logs, and driver invocations of the actual production runs that produced the released 19 866 solutions and the 1 076-trial statistics are not supplied. These process-level claims should be read as the authors' record of the search that was run, not as independently reproducible certificates in the same sense as the edge sets. Worse, as detailed in Section 7's Diversification discussion below, we found on direct execution that the released, unmodified `c4free_sa.py` frequently cannot progress at all from a fresh random start under the documented parameter ranges, for a structural reason (not merely slow convergence); we could not identify parameters under which it reaches the target edge count with zero violations. A reader wishing to verify the exact process-level statistics should therefore not assume rerunning the released code as documented will reproduce them, or indeed complete a single trial successfully; see Section 7 for the specifics and for what remains an open provenance question about the actual search method used.

7.1 Two-phase simulated annealing

Phase 1 (Penalty SA). The acceptance criterion for proposed moves minimises $f(E) = -|E| + \lambda V$ where V is the number of C_4 violations and $\lambda \in [0.30, 0.90]$, but the "best" state `phase1_sa` tracks and returns is *not* the minimiser of f: the released code instead keeps the state with lexicographically smallest $(V, -|E|)$ seen so far (`v_cur < best_v or (v_cur == best_v and size_cur > best_size)`) – prioritising fewer violations first, and only using edge count as a tiebreaker among states with equal V. Temperature schedule: geometric from $T_0 \in [0.20, 4.00]$ to $T_1 \in [0.001, 0.030]$ over 3–15 million steps. A move toggles a single edge; the change in f is computed incrementally in $O(\deg)$ time using precomputed C_4-to-edge incidence lists.

Phase 2 (Swap SA). Edge count is fixed; swap moves (remove one edge, add one non-edge) minimise V. Phase 2 only runs when phase 1 *returns* a best-so-far state with $V \leq 4$ violations – `phase1_sa` tracks and returns the best state seen during the run, not necessarily its terminal current state, so a literal reimplementation using only the terminal state could behave differently. In the released code, the step budget for phase 2 is chosen by the phase-1 violation count v, not by dimension: 2×10^8 steps if $v \leq 1$, otherwise 5×10^7 steps. In practice this correlates with dimension, since phase 1 typically ends closer to $v \leq 1$ at the Q_7/305-edge target than at the Q_8/681-edge target, but the branch itself is on v, not on n or the target edge count.

Diversification. The released search code (`c4free_sa.py`) computes, before each trial, a random element of $\mathrm{Aut}(Q_n)$ (random bit permutation composed with random bit flip) applied to the current best solution, intended as a diversified restart point. However, this computed value is not passed into `phase1_sa`, which unconditionally initialises from a fresh uniform-random edge set of the target size (`rng.sample(range(ne), target)`) regardless of any prior best solution; the automorphism-based restart is dead code in the released implementation. (The script's own docstring originally described this mechanism without noting the bug; we have appended a note to the docstring documenting it, without altering any functional code, so that a reader of the code alone is not misled.)

A more serious defect in `phase1_sa`. On direct execution, we found that the released `phase1_sa` is frequently unable to make *any* progress at all, for a structural reason independent of step count. Each accepted or rejected proposal toggles a single edge, changing the violation count V by at most $n-1$ (the number of squares containing that edge); a proposal is rejected outright whenever the resulting V would exceed `viol_cap` (drawn per trial from $[6, 40]$). If the initial random edge set's V exceeds `viol_cap` $+ (n-1)$, *no* single-edge toggle can ever produce an accepted move, so the state is frozen from step one, for any number of steps. We verified this is not a rare occurrence but the typical case: reproducing the exact RNG state of trial 1 for `--n 7 --target 304 --seed 42`, the initial violation count is $V = 152$ against `viol_cap` $= 14$ (best single-toggle result: 147, still far above cap), and running `phase1_sa` for 100 000 steps leaves $V = 152$ unchanged. Reproducing trial 1 of `--n 8 --target 680 --seed 0` similarly gives initial $V = 333$ against `viol_cap` $= 38$, frozen after 50 000 steps. Sampling 10 000 independent uniform-random edge sets directly (bypassing the annealing loop) gives a minimum V of 117 for $Q_7/304$ and 297 for $Q_8/680$ – both far above the maximum possible `viol_cap` of 40 – so this is not attributable to unlucky seeds. We could not identify parameter draws under which the released, unmodified `phase1_sa` reaches $V = 0$ from a fresh random start for these targets. Consequently, our earlier claim that fresh per-trial random initialisation explains the diversity of the released solutions should be read with this caveat: we can confirm the released *solutions* are valid (independently verified C_4-free certificates), but we cannot confirm that the released, unmodified `phase1_sa` – run as literally documented – is what actually produced them. Either the historical production code differed from what is released here (e.g. a different `viol_cap` range or a different acceptance rule), or some other undocumented step was involved; we do not know which, and we did not attempt to guess and silently "fix" the released code, since that would misrepresent what was actually run. This is a genuine, unresolved gap in the provenance of the search *method* (as distinct from the search *outputs*, which remain independently verified).

The diversity actually observed across the 19 866 collected solutions is therefore not fully explained by the mechanisms described in this section; we note for completeness that even had the intended automorphism-based restart mechanism been wired correctly, applying an automorphism to a solution does not change which $\mathrm{Aut}(Q_n)$-orbit it belongs to, and the annealing moves used here (uniform edge toggle/swap proposals against an automorphism-invariant objective) admit an $\mathrm{Aut}(Q_n)$-equivariant Markov kernel; any diversification benefit would have come only from decorrelating the specific labelled trajectory taken by a finite stochastic run; it does not diversify the orbit sampled from.

7.2 Comparison with known lower bounds

Brass–Harborth–Nienborg's weaker bound is stated for $n \geq 9$; the values below evaluate that formula at $n = 7, 8$ as an extrapolation for illustrative comparison, not as a proven bound at those dimensions (see Table 1 and the Remark in Section 4).

$$n = 7: \quad \tfrac{1}{2}(7 + 0.9\sqrt{7}) \cdot 64 \approx 300.2 \text{ (extrapolated)}, \quad \text{bound: } 304\ (\Delta = +3.8,\ +1.27\%),$$
$$n = 8: \quad \tfrac{1}{2}(8 + 0.9\sqrt{8}) \cdot 128 \approx 674.9 \text{ (extrapolated)}, \quad \text{bound: } 682\ (\Delta = +7.1,\ +1.05\%).$$

The $n = 8$ figure is the odd-square reconstruction of Section 5; the simulated-annealing method of this section independently reaches 680 edges on Q_8 ($\Delta = +5.1$, $+0.75\%$; Section 4).

7.3 Regularity trend across dimensions

Degree sequences across dimensions: Q_6: $\{4^{56}, 5^8\}$; Q_7: $\{4^{32}, 5^{96}\}$; Q_8: $\{4^8, 5^{160}, 6^{88}\}$ (Solution A; Section 4). The minimum degree is 4 in all three cases; the degree support is $\{4, 5\}$ for both Q_6 and Q_7 (unchanged from $n = 6$ to $n = 7$) and widens to $\{4, 5, 6\}$ only at Q_8. Beyond this, the exponents do not follow an evident closed-form pattern: Q_6's $\{56, 8\}$ split is markedly more skewed than Q_7's $\{32, 96\}$ or Q_8's $\{8, 160, 88\}$, so we do not claim a general regularity trend here.

8 Computational Reproduction of $\mathrm{ex}(Q_6, C_4) = 132$

Harborth and Nienborg [8] proved $\mathrm{ex}(Q_6, C_4) = 132$ by combinatorial arguments. We reproduce the lower-bound (construction) side fully independently and computationally (Section 8.1); for the upper-bound side we give an ILP formulation whose feasible value matches 132 (Section 8.2), but we did not independently verify its optimality within a practical runtime, so the upper bound $\mathrm{ex}(Q_6, C_4) \leq 132$ itself rests on [8], not on our own solve.

8.1 Lower bound: explicit construction

Simulated annealing (Section 7) finds a C_4-free subgraph of Q_6 on 132 edges, certified by exhaustive enumeration of all 240 four-cycles. The explicit edge list is provided in `q6_edges_132.jsonl`.

8.2 Upper bound: integer linear program

Label the 192 edges of Q_6 as $x_0, \ldots, x_{191} \in \{0, 1\}$ (matching the released `q6_ilp.mps`'s 0-indexed variable names; the text elsewhere in this paper is otherwise 1-indexed for readability, but the MPS file itself uses 0-indexing throughout), and let $\mathcal{C}_6$ denote the set of all 240 four-cycles of Q_6. For each four-cycle $\{e_1, e_2, e_3, e_4\} \in \mathcal{C}_6$, the constraint $x_{e_1} + x_{e_2} + x_{e_3} + x_{e_4} \leq 3$ ensures C_4-freeness. The ILP is:

$$\max \sum_{i=0}^{191} x_i \quad \text{subject to} \quad \sum_{e \in C} x_e \leq 3 \ \ \forall C \in \mathcal{C}_6, \quad x_i \in \{0, 1\}.$$

This formulation has 192 binary variables and 240 constraints, each variable appearing in exactly 5 of the 240 square constraints (matching Q_6's edge-square incidence). The MPS file `q6_ilp.mps` is provided for independent verification (SHA-256 in the released checksum manifest). The correspondence between each x_i and its Q_6 edge was not originally recorded alongside the MPS file. We have since constructed an incidence-preserving identification (`q6_ilp_edge_map.csv`: $x_i \leftrightarrow$ vertex pair, verified to reproduce all 240 constraints' variable memberships exactly from the released MPS file) by matching the MPS's variable–constraint incidence pattern against Q_6's known edge–square incidence structure under one specific, standard enumeration convention (integer vertex labels $0, \ldots, 63$; squares enumerated by base vertex, then direction pair). We call this an identification rather than *the* original correspondence because incidence structure alone determines it only up to $\mathrm{Aut}(Q_6)$: any automorphism compatible with the assumed labelling convention would give an equally valid, incidence-matching relabelling. The identification is nonetheless useful for interpreting individual solution vectors and is consistent with (not merely coincidentally matching) the released data throughout.

Proposition 16. *The ILP above has feasible value* 132*, matching the released 132-edge construction (Section 8.1); combined with Harborth and Nienborg's independent combinatorial proof that* $\mathrm{ex}(Q_6, C_4) \le 132$ *[8], this gives* $\mathrm{ex}(Q_6, C_4) = 132$*. We did not independently verify optimality of* 132 *for this ILP instance within a practical runtime (see below); the upper bound* $\mathrm{ex}(Q_6, C_4) \le 132$ *used here rests on [8], not on our own solve of the ILP.*

We independently re-solved `q6_ilp.mps` with HiGHS 1.15.1 (Python bindings via `highspy`; default settings, single thread, single-threaded x86_64 Linux VM, Intel Xeon @ 2.10GHz, 4GB RAM; invocation: `Highs().readModel(...)` then `run()` with `time_limit` set to 60 and 260 seconds respectively). A feasible solution matching the released 132-edge construction was found quickly, but the solver did not close the optimality gap within several minutes: after 60s the gap was 6.82% (dual bound -141), and after 260s it was 6.06% (dual bound -140), against the primal bound of -132 throughout. As these figures are implementation- and machine-dependent, they should be read as evidence that the gap does not close quickly under a generic default-settings solve, not as a precise benchmark. This is consistent with the large automorphism group of Q_6 ($|\mathrm{Aut}(Q_6)| = 6! \cdot 2^6 = 46\,080$) producing many symmetric alternative optima, which is known to slow generic branch-and-bound without dedicated symmetry-breaking. We do not claim generic MIP solvers verify this instance's optimality quickly; the 132-edge value should be regarded as independently established by Harborth–Nienborg's combinatorial proof [8], with the ILP providing a machine-checkable feasible witness for the primal side and a formulation that a sufficiently long or specialised solve can, in principle, verify on the dual side.

9 Open Problems

1. Prove or disprove $\mathrm{ex}(Q_7, C_4) = 304$ (Conjecture 15).
2. Determine $\mathrm{ex}(Q_8, C_4)$ exactly. Is it 682, or can a non-odd-square construction exceed $g(8) = 682$? The Q_7 audit of Section 6.4 shows the two questions are genuinely different for $n = 7$ (the general extremal value 304 is attained overwhelmingly by non-odd-square solutions), so no assumption is made here about which way $n = 8$ resolves.
3. What is the number of $\mathrm{Aut}(Q_7)$-orbits among all 304-edge C_4-free subgraphs of Q_7? Among the 389 odd-square solutions specifically?
4. Is the degree sequence $\{4^{32}, 5^{96}\}$ necessary for any locally maximal C_4-free subgraph of Q_7 on 304 edges?
5. Can the flag algebra method of [1, 2] yield tight bounds for small $n \le 8$?

6. Does the degree support of locally maximal C_4-free subgraphs of Q_n near the extremal edge count widen beyond $\{4, 5\}$ again for $n \geq 9$ (as it did once, from Q_7 to Q_8; Section 7.3), and is minimum degree 4 necessary at these edge counts for $n \geq 9$?

7. We now know $g(6) = \mathrm{ex}(Q_6, C_4) = 132$ (Theorem 2 combined with [8]). For $n = 7, 8$, Theorem 2 only establishes $g(7) = 304$ and $g(8) = 682$; whether $\mathrm{ex}(Q_7, C_4) = g(7)$ and $\mathrm{ex}(Q_8, C_4) = g(8)$ remains exactly as open as Conjecture 15 and Open Problem 2 above, since $\mathrm{ex}(Q_n, C_4)$ itself is not known exactly for $n = 7, 8$. (The coincidence at $n = 6$ is in any case not explained by the odd-square construction alone for $n = 7$: only 389 of the 19 866 known 304-edge Q_7 solutions are odd-square.) Does $g(n) = \mathrm{ex}(Q_n, C_4)$ continue to hold for $n = 9, 10, \ldots$ (once $\mathrm{ex}(Q_n, C_4)$ is itself determined), or does the gap $\mathrm{ex}(Q_n, C_4) - g(n)$ become positive at some n? Laplante et al. [9] give general lower and upper bounds on the frustration index $|E(Q_n)| - g(n)$ that match only when n is a power of 4; whether $g(n) = \mathrm{ex}(Q_n, C_4)$ persists beyond $n = 8$ is, in their language, a question about when this frustration-index gap and the C_4-free extremal gap coincide.

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Use of generative AI. The author used Claude (Anthropic) for: (i) language editing of the abstract; (ii) assistance in writing the supplementary verification script (`verify.py`) used to confirm C_4-freeness of the released edge-list certificates; and, for this revision, (iii) retrieval and cross-checking of the primary sources cited in Section 5 and (iv) drafting assistance for the revised text. All definitions, constructions, theorems, computations, and analyses are the author's own, including the 256-bit spin configuration used in the Q_8 witness of Section 5.3, whose provenance is recorded there as a dated correspondence account. The author reviewed all AI-assisted output, independently executed the verification and witness scripts against the released data, and takes full responsibility for the accuracy and integrity of the work. The specific model version(s) of Claude used across the original submission and this revision were not recorded at the time and cannot be confirmed retrospectively; the disclosure above is accurate as to which tasks AI assistance was used for, but not as to the exact model version for each.

Disclosure statement

No potential conflict of interest was reported by the author.

References

[1] R. Baber, Turán densities of hypercubes, arXiv:1201.3587, 2012.

[2] J. Balogh, P. Hu, B. Lidický, and H. Liu, Upper bounds on the size of 4- and 6-cycle-free subgraphs of the hypercube, *Eur. J. Combin.* **35** (2014), 75–85.

[3] P. Brass, H. Harborth, and H. Nienborg, On the maximum number of edges in a C_4-free subgraph of Q_n, *J. Graph Theory* **19** (1995), 17–23.

[4] F. Chung, Subgraphs of a hypercube containing no small even cycles, *J. Graph Theory* **16** (1992), 273–286.

[5] B. Derrida, Y. Pomeau, G. Toulouse, and J. Vannimenus, Fully frustrated simple cubic lattices and the overblocking effect, *J. Physique* **40** (1979), 617–626.

[6] M. R. Emamy-K., P. Guan, and P. I. Rivera-Vega, On the characterization of the maximum squareless subgraphs of the 5-cube, *Congr. Numer.* **88** (1992), 97–109.

[7] P. Erdős, On some problems in graph theory, combinatorial analysis and combinatorial number theory, in: *Graph Theory and Combinatorics* (B. Bollobás, ed.), Academic Press, 1984, pp. 1–17.

[8] H. Harborth and H. Nienborg, Maximum number of edges in a six-cube without four-cycles, *Bull. ICA* **12** (1994), 55–60.

[9] S. Laplante, R. Naserasr, A. Sunny, and Z. Wang, Sensitivity conjecture and signed hypercubes, *ACM Trans. Comput. Theory* **18** (2026), no. 1, Article 6, 1–25, https://doi.org/10.1145/3777401 (preprint: HAL hal-04080411, 2023).

[10] E. Marinari, G. Parisi, and F. Ritort, The fully frustrated hypercubic model is glassy and aging at large D, *J. Phys. A* **28** (1995), 327–334, arXiv:cond-mat/9410089.

[11] A. Thomason and P. Wagner, Bounding the size of square-free subgraphs of the hypercube, *Discrete Math.* **309** (2009), 1730–1735.